

with high-bandwidth requirements, such as high-definition video, 3-D action gaming and mobile projection. DRAMs equipped with the technology are expected to arrive in the second half of 2011.

Rambus' Mobile XDR, another incumbent, is capable of achieving data rates of 3.2 to 4.3 Gbps and is suited for next-generation smart phones, netbooks and mobile gaming and multimedia products.

Sturdy, steady interfaces for your storage needs

In the storage space too, the interface used by a device would vary depending on its application. Storage devices like thumb drives would have interfacing needs different from that of enterprise storage servers.

In general, the serial advanced technology attachment (SATA) and serial attached SCSI (SAS) have emerged as the most prominent industry standards for storage devices. SATA and SAS are serial technologies that replace the erstwhile parallel ATA and SCSI protocols (see the box for explanation on the rising popularity of serial technologies). While SATA is used in optical drives, hard drives, port multipliers and bridge devices, SAS is primarily used in enterprise storage applications.

First- and second-generation SATA provide transfer capacities of 1.5 Gbps and 3 Gbps, respectively, while the latest versions (SATA 3.0 and SAS 2.0) operate at a whopping 6 Gbps.

External storage devices like thumb drives and portable hard disks use external interfaces like USB, FireWire or external-SATA (eSATA). "USB 2.0 is widely-used and is almost a universal interface. Its successor USB 3.0, which offers transfer speeds of around 400 MBps, is also rapidly gaining popularity. The FireWire 400/800 interfaces deliver isochronous transfer of data, and enable power supply just like USB. Devices featuring the eSATA interface also offer fast transfer speeds and are also more affordable. However, the eSATA interface is not suitable for

Serial versus parallel interfaces

Interfaces can be broadly divided into serial and parallel standards. Serial technologies transfer the data one bit at a time over a communication channel or computer bus. Parallel interfaces transmit multiple bits at a time. So while it would take eight clock cycles to transmit a byte of information using serial technologies, parallel interfaces would take just one clock cycle to achieve the same task.

Although parallel interfaces seem faster and more desirable at the outset, the fact is that many vendors nowadays prefer serial to parallel interfaces. This is because the synchronisation of bit transfer at higher data rates and clock frequencies becomes problematic in the case of parallel transmission, which can also not handle large distances easily. Moreover, serial technologies have now become much faster and less complex, and in many ways outdo the benefits of parallel technologies while overcoming their shortcomings! Several new serial data bus architectures including USB, PCI Express, XAUI, RapidIO, HDMI and SATA provide a significantly greater data throughput than possible a few years ago. Serial standards are not flawless either—many a times, they add to the design complexity to an extent that could increase the cost of development and time-to-market if not handled properly. Nevertheless, the benefits make the efforts worthwhile.

Evan Sun, technical marketing manager, Tektronix-Asia Pacific Region, comments, "For sure, every serial standard is entering the third generation in the light of the rapid increase in real-time experience. DDR3 memory has been widely deployed in not only traditional computers but also graphic cards, handheld devices and smart phones. USB 3.0 at 5 Gbps is now becoming a standard I/O interface in the computer segment, being ten times faster than the previous USB 2.0 at 480 Mbps. Due to the high storage capacity requirements, SATA 3G and SAS 2G at 6 Gbps are now used in both computer and server segments. In general-purpose I/O interconnects, PCI Express has been playing a bigger role for a wide variety of computing and communication platforms. Moreover, PCI Express Gen3 at 8 GT/s is expected to improve data output significantly. Of course, we see 10 Gbps, 40 Gbps and 100 Gbps becoming hot technologies in the Ethernet market."

Some of today's high-speed interfaces include DDR2 and DDR3 in the memory space; PCIe Gen2, USB2 and USB3 in the input/output space; 802.11n in the wireless communications space; and HDMI in the high-definition multimedia space

end-users as it cannot supply power to the device," says Khalid Wani, sales director-Branded Products, India, Middle East & Africa, Western Digital. A workaround has been developed, in the form of the Power over eSATA port (eSATAp), which combines the speed of eSATA with the compatibility of USB.

USB 3.0—perhaps one of the

most significant developments last year—is by far the fastest interface available for external drives. It offers a theoretical maximum rate of 5 Gbps, which is nearly twice as fast as the eSATA interface (3Gbps) and six times faster than FireWire 800 (800 MBps). In practical usage, a flash drive with USB 3.0 can transfer 1 GB of data in just over three seconds, compared to 33 seconds with a USB 2.0 interface. That is like transferring three movies in less than ten seconds!

Jack of all trades

USB is not just for storage. It is a general-purpose input-output (I/O) interface that can easily be claimed as one of the most popular ones in this category! You might be interested in knowing that this highly-compatible and fast interface, which has now replaced serial and parallel ports in most PCs, was developed by an Indian named Ajay Bhatt when he was at In-